BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Network Attacks

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. How does phishing exploit users?

2. A star network's central switch fails. What happens?
   A) Only one cable fails
   B) All files are deleted
   C) Connected devices lose contact
   D) It becomes a mesh

3. Why can a mesh topology be resilient?

4. (a) Convert binary 10110110 to hexadecimal, using four-bit groups. (b) Convert hexadecimal B6 into eight-bit binary.

Today's objectives

Recognise brute-force, DoS/DDoS, interception and SQL injection. Explain the effect of each attack on a system.

Key vocabulary - Read, cover, recall

Brute-force attack
Repeated guessing of possible passwords.

DoS
Denial of Service; overloads availability of a service.

DDoS
Distributed Denial of Service; traffic from multiple sources.

Data interception
Capturing information while it is transmitted.

BLACK PEN: Write one definition in your own words.

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New Learning - Key ideas

Brute-force guessing

An attacker systematically tries many possible passwords in an attempt to gain access to an account.

Denial of Service

A DoS attack overwhelms a service with traffic or requests so legitimate users cannot access it. A DDoS attack originates from multiple systems.

Data interception

Data may be captured during transmission, potentially exposing confidential information if not protected.

SQL injection

A malicious input may be interpreted as part of a database query, causing unauthorised viewing or alteration of data if the application handles input insecurely.

I Do - Worked model

A school's web service receives overwhelming requests and genuine users cannot log in. Explain the likely attack and its effect. [4 marks]

- The service is overloaded and genuine users cannot access it.
- A denial-of-service (DoS) attack targets service availability [1].
- Excess traffic or requests [1] overload server resources [1].
- Legitimate pupils and staff are prevented from accessing the website [1].

We Do - Guided practice

1. An account receives thousands of unsuccessful login attempts. Explain the attack and its likely aim. [4 marks]

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2. A user sends private details on an unprotected connection. Explain data interception and a consequence. [4 marks]

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3. An insecure database-backed website misinterprets crafted input as part of a query. Explain the threat and possible damage. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write the final line that shows the number of failed attempts.

```
failed = 2
failed = failed + 1
```

You Do - Independent exam questions

Q2. Define a brute-force attack. [2 marks]

Q3. A museum's online ticket-booking service is deliberately overwhelmed with requests. Explain TWO consequences of the denial-of-service attack for the museum and its visitors. [4 marks]

Q4. Describe DoS, interception and SQL injection, including an effect of each. [6 marks]

Exit Ticket - True or False?

A DoS attack aims to make a service unavailable to legitimate users.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Phishing pretends to be a trusted source to trick people into sharing data or doing something unsafe.

2. C - Connected devices lose contact

3. Other paths can keep data moving if some links fail. More links add cost and complexity.

4. (a) B6: 1011 0110. (b) 10110110.

I Do - Explain the model

The service is overloaded and genuine users cannot access it.

A denial-of-service (DoS) attack targets service availability [1].

Excess traffic or requests [1] overload server resources [1].

Legitimate pupils and staff are prevented from accessing the website [1].

We Do - Model answers

1. A brute-force attack [1] tries many possible passwords [1] seeking the correct combination [1] to gain unauthorised access [1].

2. Data interception captures data during transmission [1] without authorisation [1]. Private details may be read [1] and misused for impersonation or theft [1].

3. SQL injection occurs when unsafe user input changes a database query [1] so unintended commands are executed [1]. This may reveal private records [1] or alter/delete stored data [1].

Code First - Answer

failed = 2
failed = failed + 1
print(failed)
Expected output: 3

You Do - Self-mark

Q2: Attempts many possible passwords [1] to discover the correct one/gain access [1].

Q3: The booking website becomes overloaded or unavailable [1], so genuine visitors cannot buy or view tickets [1]. The museum may lose bookings/revenue or have increased support work [1], disrupting museum operations [1].

Q4: DoS overloads a service [1], causing legitimate access failure [1]. Interception captures data in transit [1], risking disclosure [1]. SQL injection manipulates database queries via unsafe input [1], potentially revealing or changing stored records [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.